

Retail Sales Analytics & Business Intelligence Dashboard

■ Project Overview

This project presents a comprehensive Retail Business Intelligence and Sales Performance Analysis system designed to transform retail sales data into meaningful business insights.

The project focuses on sales performance, customer behavior, product performance, operational efficiency, and business decision-making using data analytics and visualization techniques.

■ Objectives

- Analyze retail sales performance and revenue trends
- Identify important business KPIs
- Understand customer purchasing behavior
- Analyze product performance
- Monitor operational efficiency
- Support data-driven business decisions
- Present insights through interactive dashboards and visualizations

■■ Technologies Used

- Python
- Power BI
- DAX
- SQL
- Data Analytics
- Data Visualization
- Machine Learning / Predictive Analytics

■ Key KPIs

KPI	Result
Total Sales	■12,50,000
Total Orders	2,450
Average Order Value	■510
Sales Growth	+12%
Cancellation Rate	8%
Average Delivery Time	3–5 Days

■ Dashboard

The dashboard presents key retail performance indicators using interactive business intelligence visualizations.

Dashboard Features

- KPI Cards
- Sales Performance Analysis
- Sales Growth Analysis

- Order Analysis
- Delivery Performance
- Cancellation Rate
- Interactive Filters / Slicers
- Business Insights

■ Analysis Performed

Sales Analysis

Analyzed total sales, order volume, average order value and sales growth to understand overall business performance.

Customer Analysis

Analyzed customer purchasing patterns and behavior to support better customer understanding and segmentation.

Operational Analysis

Analyzed delivery performance and cancellation rate to identify operational improvement areas.

Visualization

Created business intelligence visualizations including KPI cards, line charts, pie charts, clustered column charts, clustered bar charts, waterfall charts and slicers.

■ DAX Measures

Example measures implemented in Power BI:

```
Total Sales = SUM('Table'[Net Amount (■)])
```

```
Total Orders = COUNT('Table'[Order ID])
```

```
Avg Order Value = DIVIDE([Total Sales], [Total Orders], 0)
```

```
Delivery Days = DATEDIFF('Table'[Order Date], 'Table'[Delivery Date], DAY)
```

Additional measures include Sales Growth %, Cancellation Rate and Running Sales.

■ Key Insights

- The reported total sales reached ■12.50 lakh.
- The project recorded 2,450 orders.
- Average order value was ■510.
- Sales showed approximately 12% positive growth.
- Average delivery time was reported between 3 and 5 days.
- The 8% cancellation rate indicates an area for operational improvement.

■ Future Scope

- Real-time sales data integration
- Advanced machine learning models
- Improved demand forecasting
- Customer behavior prediction
- Product recommendation systems
- Cloud deployment
- Mobile dashboard access
- AI-based decision support
- Chat-based analytics
- Multi-channel retail data integration

■■■ Project Type

Academic / Data Analytics / Business Intelligence Project

■ Project Documentation

The complete project report is available in this repository as:

"Retail sales analysis.pdf"

■ Project Team

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■ Department

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